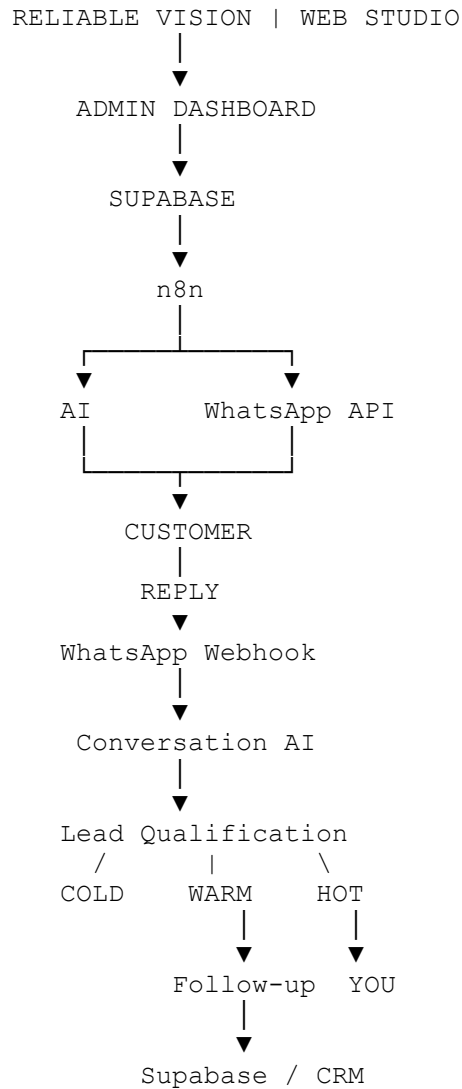


Your final system will be:



Supabase provides PostgreSQL, Auth, Storage, Realtime and APIs, so it is suitable as the central backend for this application.

PART 1 — What we are building

Your business information will be stored in the system:

Business:
Reliable Vision | Web Studio

Services:
• Business Websites

- Landing Pages
- Portfolio Websites
- Website Redesign

Location:
Tamil Nadu, India

Website:
<https://santhosh-portfolio-gamma.vercel.app/>

Email:
santhosh.rv.work@gmail.com

WhatsApp:
+91 9597482991

The AI will use this information.

PART 2 — The tools

We will use:

Tool	Purpose
Antigravity	Build the dashboard/application
Next.js + TypeScript	Website
Tailwind CSS	UI
Supabase	Database + authentication
n8n	Automation engine
AI API	Personalization + conversation + qualification
WhatsApp Cloud API	Customer messaging
Google Sheets	Optional CRM/export
GitHub	Code/version control
Vercel	Dashboard deployment

n8n supports self-hosting through npm/Docker as well as cloud hosting, and is specifically designed to connect applications and APIs into workflows.

PART 3 — VERY IMPORTANT: don't connect WhatsApp yet

Do **not** start with WhatsApp.

Do **not** create API keys yet.

Do **not** pay for anything yet.

First we'll make:

```
Admin Dashboard
  ↓
Supabase
  ↓
Excel Upload
  ↓
Leads stored correctly
```

Once that works, we'll add n8n.

Then AI.

Then WhatsApp.

This prevents a huge mess when debugging.

STEP 1 — Create the project

Open **Antigravity**.

Create a new project/repository:

```
reliable-vision-automation
```

I recommend this name rather than starting with a generic name because this is specifically for your business.

STEP 2 — Create GitHub repository

Create a **private** GitHub repository:

```
reliable-vision-automation
```

Don't upload API keys or passwords to GitHub.

STEP 3 — Create Supabase

Create a Supabase project:

Project name:
Reliable Vision Automation

Use the free/development plan initially.

Supabase gives each project a Postgres database and provides APIs, authentication, storage and other backend services.

Don't manually create random tables yet.

We'll create the proper schema.

STEP 4 — Give Antigravity the master project instruction

Open your project in Antigravity.

Give it this **exact prompt**:

You are the senior software architect and full-stack engineer for this project.

PROJECT NAME:
Reliable Vision AI Lead Automation

BUSINESS:
Reliable Vision | Web Studio

BUSINESS DESCRIPTION:
Professional Website Development Services.

SERVICES:
- Business Websites
- Landing Pages
- Portfolio Websites
- Website Redesign

SERVICE AREA:
Tamil Nadu, India

PORTFOLIO:

<https://santhosh-portfolio-gamma.vercel.app/>

BUSINESS EMAIL:

santhosh.rv.work@gmail.com

BUSINESS WHATSAPP:

+91 9597482991

PROJECT OBJECTIVE:

Build a production-ready AI-powered lead generation, WhatsApp conversation and lead qualification platform.

The system must eventually support:

1. Secure Admin Login
2. Admin Dashboard
3. Lead management
4. Excel/CSV upload
5. Lead validation
6. Phone number validation
7. Duplicate detection
8. Supabase PostgreSQL database
9. Campaign management
10. Campaign start/pause/stop
11. n8n automation
12. AI message personalization
13. WhatsApp Cloud API
14. WhatsApp webhook
15. Customer conversation history
16. AI conversation agent
17. Lead qualification
18. Cold/Warm/Hot classification
19. Automated follow-ups
20. Hot lead notification
21. Human handoff
22. Google Sheets/CRM integration
23. Analytics
24. Settings
25. Business AI configuration

TECHNOLOGY:

Frontend:

- Next.js
- React
- TypeScript
- Tailwind CSS

Backend:

- Supabase
- PostgreSQL
- Supabase Auth

Automation:

- n8n

Messaging:

- WhatsApp Cloud API

AI:

- API-based AI provider
- Keep AI provider integration modular so it can be changed later.

Deployment:

- Vercel for frontend
- n8n hosted separately

SECURITY REQUIREMENTS:

- Never expose secret API keys in browser/client code.
- Use environment variables.
- Use server-side API routes for secret operations.
- Implement authentication.
- Implement authorization.
- Enable and properly configure Supabase Row Level Security.
- Validate all uploaded data.
- Sanitize user input.
- Never trust frontend data.
- Never commit .env files.
- Add .env.example.
- Add proper error handling.
- Add loading states.
- Add empty states.
- Add retry handling where appropriate.
- Log important automation failures safely.
- Do not expose private customer information unnecessarily.

IMPORTANT DEVELOPMENT RULE:

DO NOT BUILD THE ENTIRE SYSTEM AT ONCE.

Build it in verified stages.

STAGE 1:

Project architecture and folder structure.

STAGE 2:

Supabase schema and security.

STAGE 3:

Admin authentication.

STAGE 4:

Admin dashboard UI.

STAGE 5:

Lead management.

STAGE 6:

Excel/CSV import and validation.

STAGE 7:
Campaign management.

STAGE 8:
n8n integration.

STAGE 9:
AI integration.

STAGE 10:
WhatsApp integration.

STAGE 11:
Webhook and conversation AI.

STAGE 12:
Cold/Warm/Hot qualification.

STAGE 13:
Follow-up automation.

STAGE 14:
CRM and analytics.

STAGE 15:
Testing, security audit and deployment.

FOR NOW:

DO NOT IMPLEMENT THE APPLICATION.

First provide:

1. Complete architecture
2. Recommended folder structure
3. Database tables
4. Database relationships
5. Authentication design
6. API design
7. Admin dashboard pages
8. n8n architecture
9. AI architecture
10. WhatsApp architecture
11. Security architecture
12. Development roadmap
13. Testing strategy

Do not invent missing business requirements.

Do not use fake APIs.

Do not create placeholder integrations that pretend to work.

Wait for approval before implementing Stage 1.

STEP 5 — What Antigravity should give you

It should produce something similar to:

```
reliable-vision-automation/
```

```
|
├── app/
│   ├── login/
│   ├── dashboard/
│   ├── leads/
│   ├── campaigns/
│   ├── conversations/
│   ├── followups/
│   ├── analytics/
│   └── settings/
├── components/
├── lib/
│   ├── supabase/
│   ├── ai/
│   └── validation/
├── api/
├── database/
│   └── migrations/
├── n8n/
├── types/
├── docs/
├── .env.example
├── .gitignore
└── README.md
```

The exact structure can differ; **don't force this exact folder tree** if Antigravity proposes a better Next.js architecture.

STEP 6 — Database design

After the architecture is approved, we'll create the database.

The important tables will be:

```
users
business_settings
```


leads
campaigns
campaign_leads
conversations
messages
followups
notifications
ai_logs

business_settings

This stores your Reliable Vision information:

business_name
description
services
service_area
website
email
whatsapp_number
ai_instructions

leads

id
name
phone
business
category
status
campaign_status
last_message_at
last_reply_at
next_followup_at
created_at
updated_at

messages

id
lead_id
conversation_id
direction
message
whatsapp_message_id
created_at

campaigns

id
name
status
start_time
end_time

```
message_template
created_at
```

followups

```
id
lead_id
scheduled_at
followup_number
status
created_at
```

This structure allows the system to remember the complete conversation instead of treating every WhatsApp message as a new customer.

Supabase recommends Row Level Security when exposing database access to the application, so we'll include RLS in the database stage rather than adding security as an afterthought.

STEP 7 — Build Admin Login

The first real screen:

```
RELIABLE VISION

AI AUTOMATION



Email
  Password
  LOGIN


```

We'll use Supabase Auth rather than creating our own password system. Supabase Auth supports email/password and other authentication methods and works with database authorization/RLS.

STEP 8 — Build Admin Dashboard

After login:

Reliable Vision AI	
Dashboard	

Leads	Total Leads	1,000
Campaigns	Contacted	500
Messages	Replies	100
Followups	 Hot	20
Analytics	 Warm	45
Settings	 Cold	435

STEP 9 — Business Settings

This is where your AI gets its knowledge.

Create:

SETTINGS

Business Name

Reliable Vision | Web Studio

Description

Professional Website Development Services

Services

☒ Business Websites

☒ Landing Pages

☒ Portfolio Websites

☒ Website Redesign

Service Area

Tamil Nadu

Website

...

Email

...

WhatsApp

...

AI Instructions

...

[SAVE]

This means later you don't have to edit code whenever you change a service.

STEP 10 — Excel Upload

Admin clicks:

Leads → Upload Excel

Excel:

Name | Phone | Business | Category

Example:

Arun | 9876543210 | Arun Textiles | Textile
Priya | 9876543211 | Priya Salon | Beauty
Ravi | 9876543212 | Ravi Motors | Automobile

The system processes:

```
Excel
↓
Read
↓
Validate
↓
Normalize phone
↓
Check duplicates
↓
Check required fields
↓
Insert valid leads
↓
Show result
```

Dashboard should show:

Import completed

Total rows:	1000
Valid:	970
Duplicates:	20
Invalid:	10

[View Leads]

STEP 11 — Now introduce n8n

Only after Excel → Supabase works.

n8n becomes your automation engine.

```
Schedule
↓
Find pending lead
↓
Lock lead
↓
Generate message
↓
Send WhatsApp
↓
Save message
↓
Update lead
```

n8n is designed to connect APIs/services and supports both cloud and self-hosted deployment.

STEP 12 — AI Personalization

The AI receives:

Name:
Arun

Business:
Arun Textiles

Category:
Textile

Your business:
Reliable Vision | Web Studio

AI generates the appropriate message.

But we will add rules:

```
Don't lie.
Don't invent services.
Don't invent prices.
Don't pretend to be human.
Don't spam.
Don't send multiple messages unnecessarily.
Keep the message short.
Use the actual business information.
```

STEP 13 — WhatsApp

Only now we configure Meta/WhatsApp.

The architecture becomes:

```
n8n
↓
WhatsApp Cloud API
↓
Customer
```

Then:

```
Customer
↓
WhatsApp
↓
Webhook
↓
n8n
↓
Supabase
↓
AI
↓
WhatsApp
```

Important: Don't delete/change your current WhatsApp Business number yet. We'll first determine the correct Meta/WhatsApp setup for that number.

STEP 14 — AI Conversation

Customer:

Hi, I need a website.

AI gets:

```
Customer profile
+
Business configuration
+
Previous messages
+
Latest message
```

Then produces:

```
reply
intent
lead_temperature
needs_human
```

For example:

```
{
  "reply": "Sure. We provide business websites, landing pages, portfolio
websites and redesign services. Could you tell me what type of business you
have?",
  "intent": "website_inquiry",
  "lead_temperature": "warm",
  "needs_human": false
}
```

OpenAI's current API supports server-side text generation through the Responses API, and API keys should be kept securely as environment variables rather than exposed in frontend code.

STEP 15 — Lead qualification

The system evaluates the conversation.

COLD

```
Not interested
No requirement
Maybe later
```

WARM

```
Tell me more
How much?
What do you offer?
I'll discuss with my partner
```

HOT

```
I want to start
Can we call?
Send proposal
I need this urgently
Let's discuss the project
```

Then:

```
HOT
↓
Save status
↓
Notify you
```

STEP 16 — Your HOT notification

You should receive:

 HOT LEAD

Name:
Arun

Business:
Arun Textiles

Category:
Textile

Requirement:
Business Website

Latest message:
"I want to discuss the website."

Status:
HOT

Now **you take over**.

The AI shouldn't continue endlessly trying to close every lead.

STEP 17 — Follow-ups

For a warm lead:

```
WARM
↓
Set next_followup_at
↓
n8n checks periodically
↓
Follow-up due?
↓
Send appropriate message
```


The exact follow-up schedule should be configurable in your Admin Panel.

STEP 18 — CRM

Your dashboard:

LEADS

Name	Business	Category	Status
Arun	Arun Textiles	Textile	 HOT
Priya	Priya Salon	Beauty	 WARM
Ravi	Ravi Motors	Auto	 COLD

STEP 19 — Analytics

Eventually:

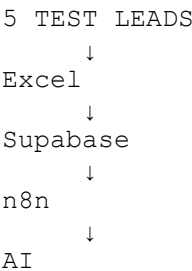
CAMPAIGN ANALYTICS

Total leads	1,000
Messages sent	800
Replies	180
Reply rate	22%
Warm leads	70
Hot leads	25
Cold leads	705

STEP 20 — Final production testing

We will not immediately upload 1,000 real customers.

First:



↓
WhatsApp
↓
Reply
↓
AI
↓
Qualification
↓
Notification
↓
Dashboard

Then:

5 → 10 → 25 → 50 → 100

Only when those tests work correctly should you scale.

The exact order we will follow

